Coil Winding Design

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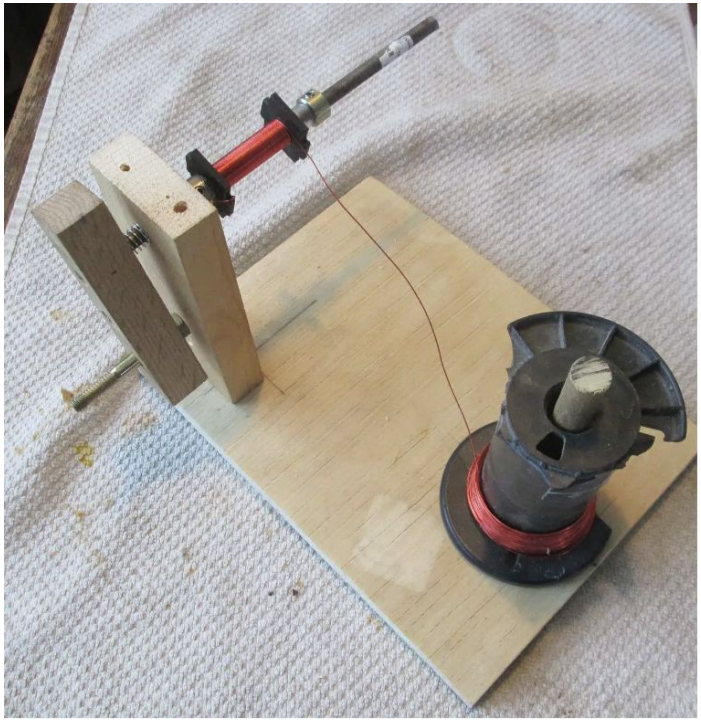
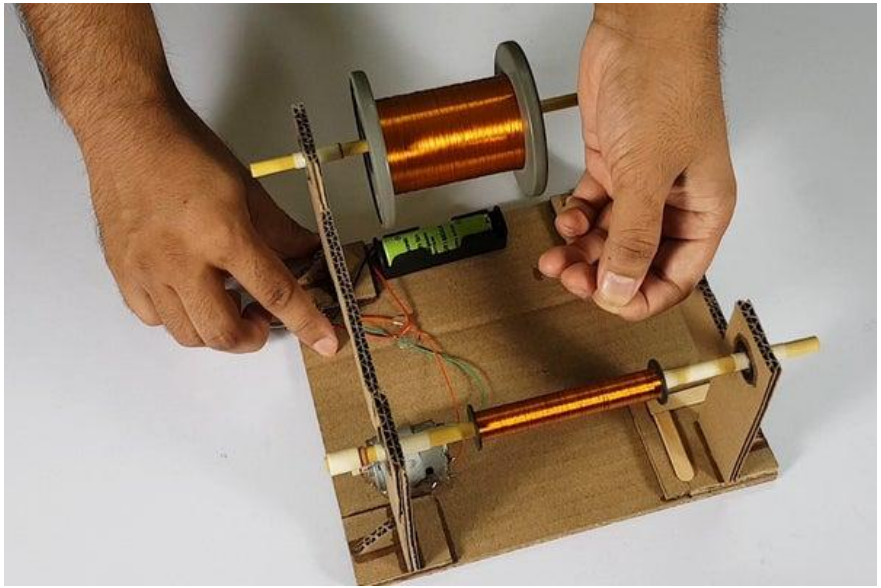
Reference design

Coil Winding Machine

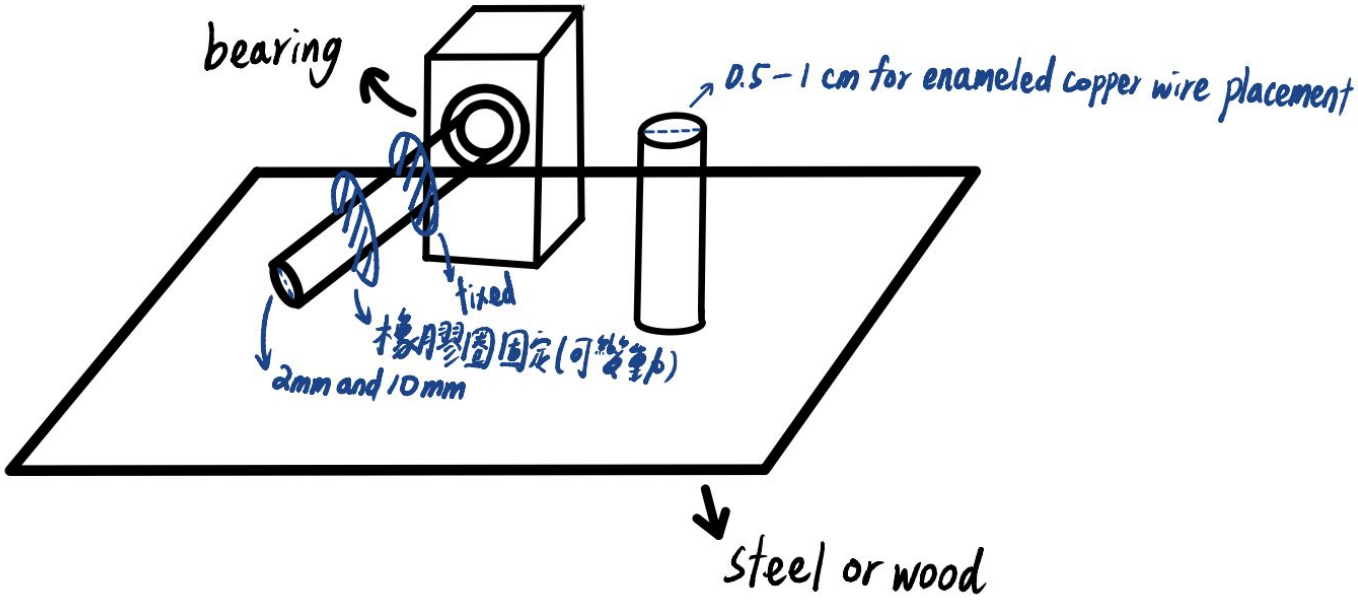




Source: AUTODESK Instructables、grwinding

Prototyping & Iteration

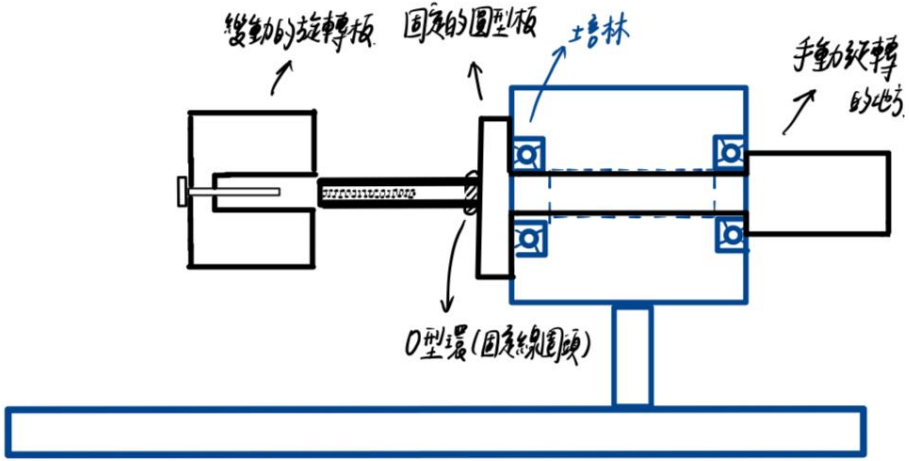
Prototype 1

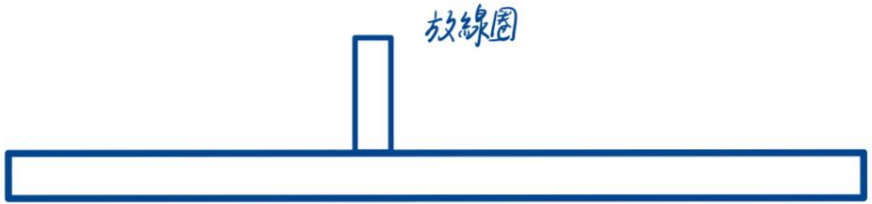

bearing
0.5 – 1 cm for enameled copper wire placement
fixed
橡膠圈固定(可發動)
2mm and 10mm
steel or wood

Prototype 1

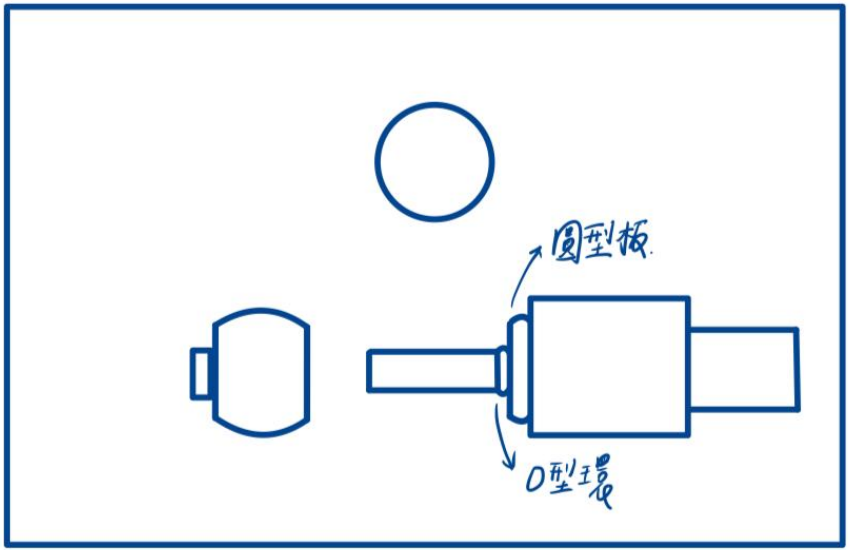
Source: Gemini

Prototype 2



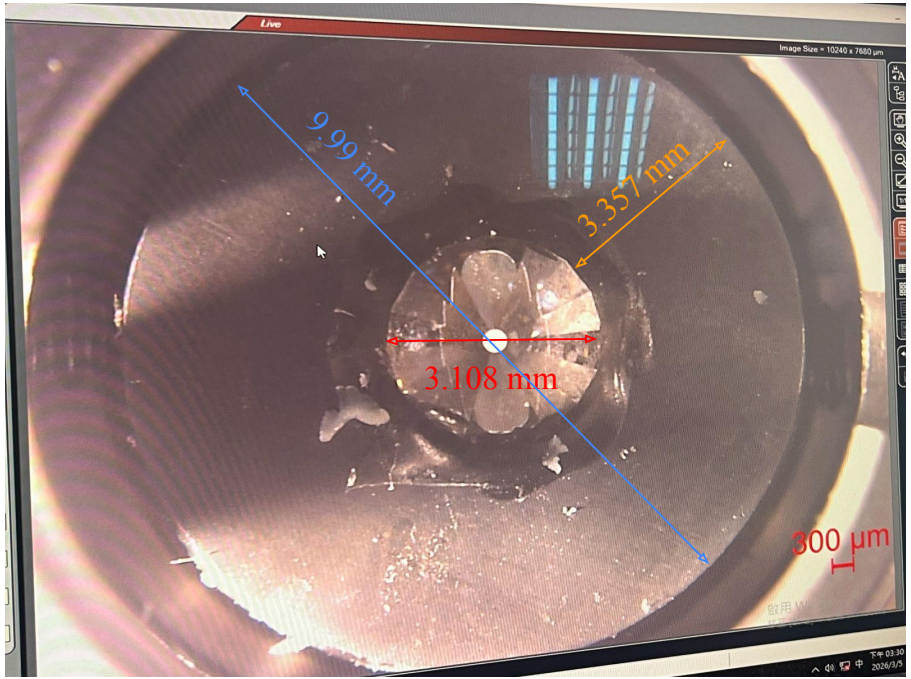


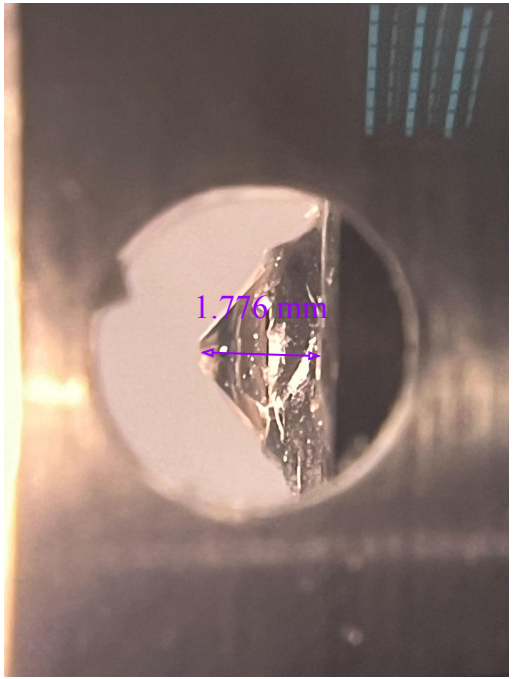
Prototype 2



Compilation of Current Diamond Anvil Cell Dimensions

DAC 154

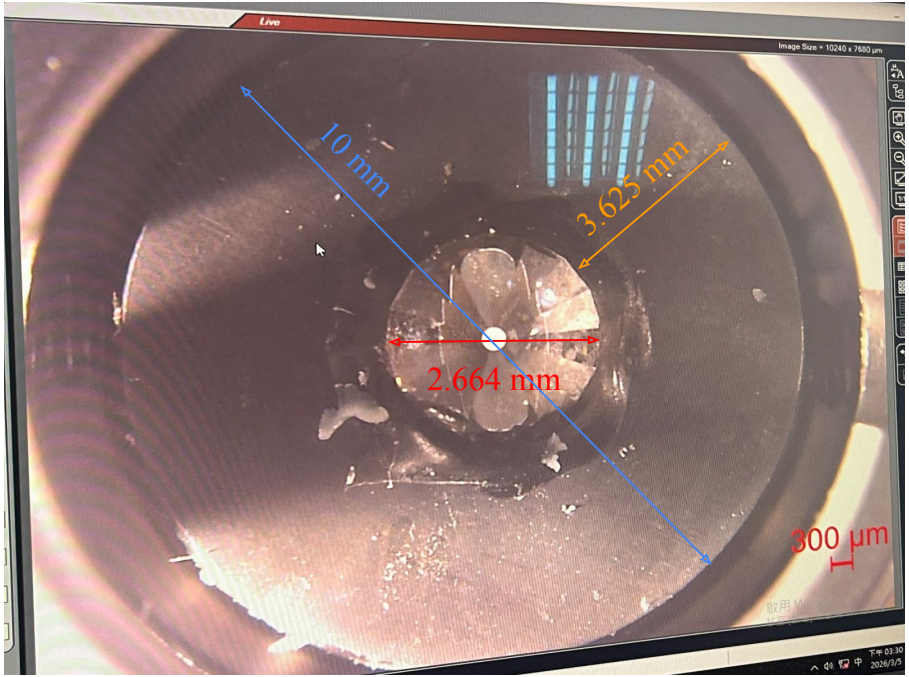


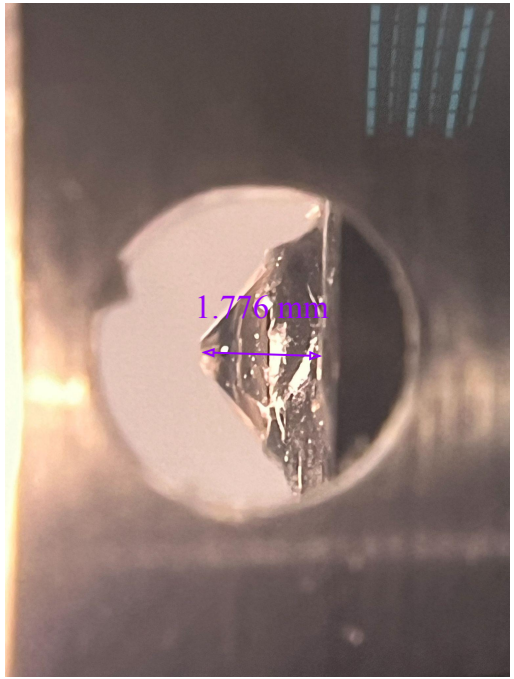


Compilation of Current Diamond Anvil Cell Dimensions

DAC 178

- An inner diameter of 9 mm is recommended for the exciting coil.



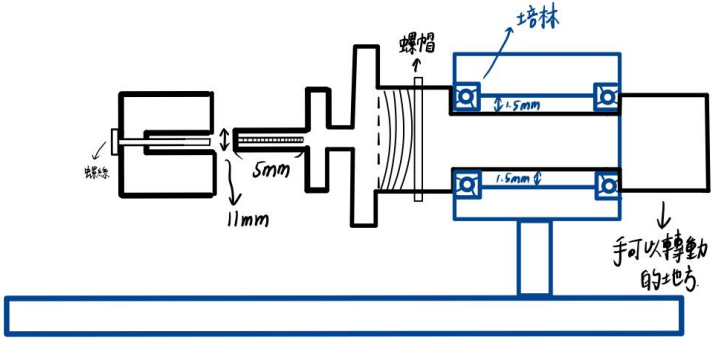


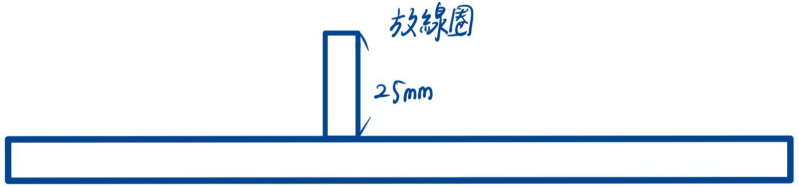
Device Size Conversion Table

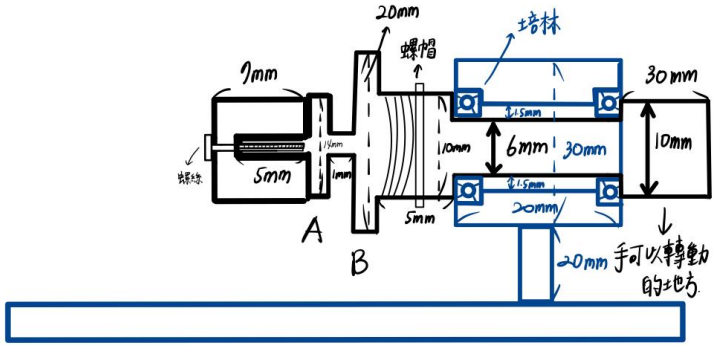
- https://docs.google.com/spreadsheets/d/1Ej7i6uHZ6vc0jknn08SA4UnEvh1v2_4u-yJeGm5euuo/edit?gid=0#gid=0
- Final selection based on DAC inner diameter and diamond thickness: Signal Coil (Ø 3 mm) and Exciting Coil (Ø 11 mm).
- Device diameter limits according to the following table: 5 mm and 14 mm, in sequence.

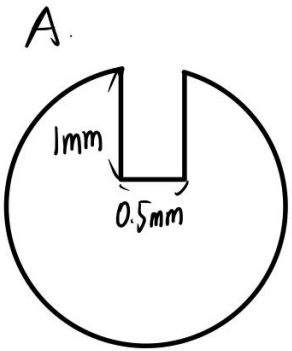
信號線圈/補償線圈				
直徑 (μm)	繞一次圓柱繞幾圈	繞幾次圓柱	厚度(mm)	直徑
15	30	6	0.45	3.18
30	30	6	0.9	3.36
40	30	6	1.2	3.48
激發線圈圈數	補償線圈圈數	訊號線圈圈數		
180	180	250		
激發線圈				
直徑 (μm)	繞一次圓柱繞幾圈	繞幾次圓柱	厚度(mm)	直徑
15	41.66666667	6	0.625	11.18
30	41.66666667	6	1.25	11.36
40	41.66666667	6	1.666666667	11.48

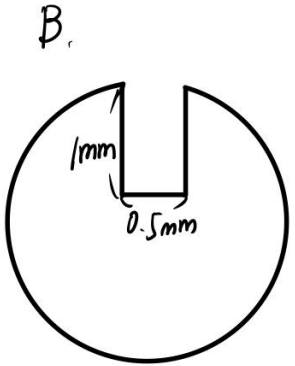
Prototype 3 (For Exciting Coil)



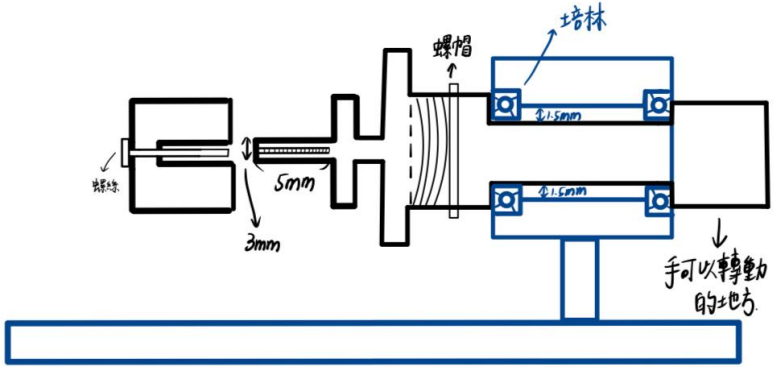


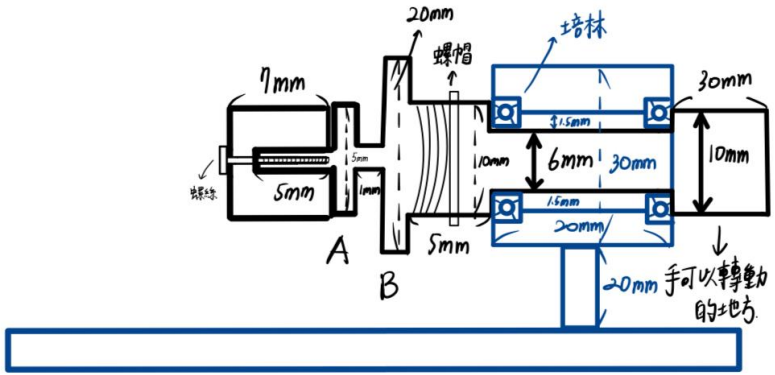




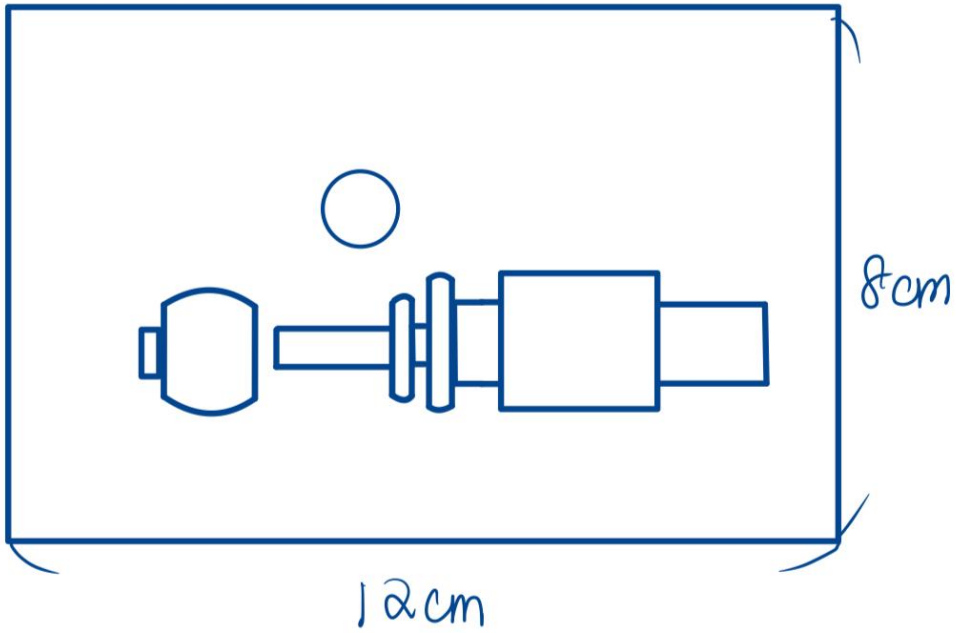


Prototype 3 (For Signal and Compensating Coil)



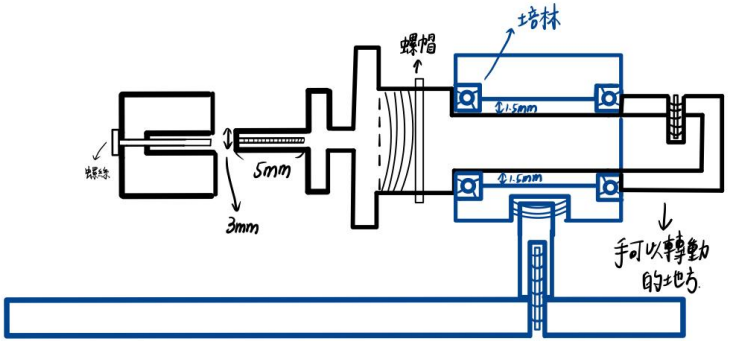


Prototype 3

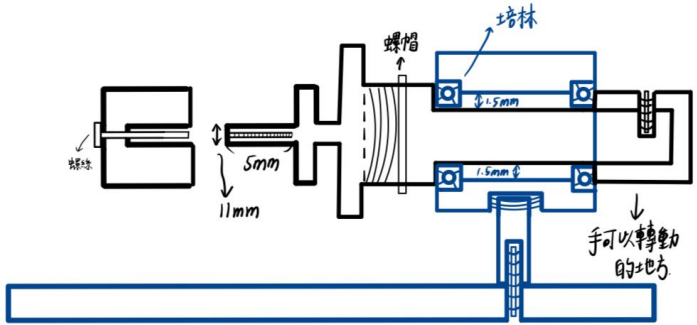


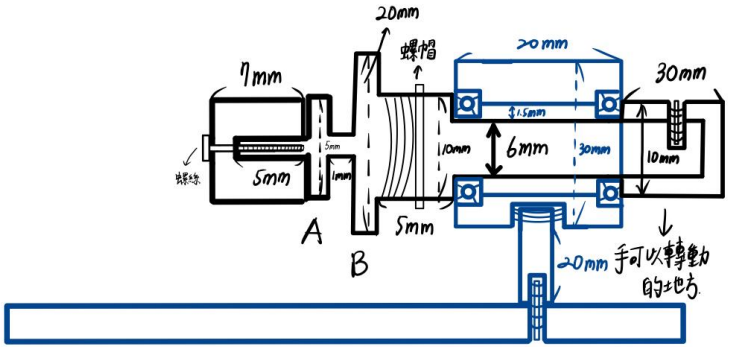
Final Design (Side View)

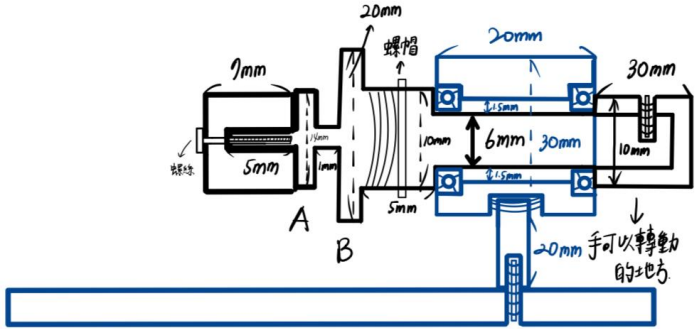
signal coil



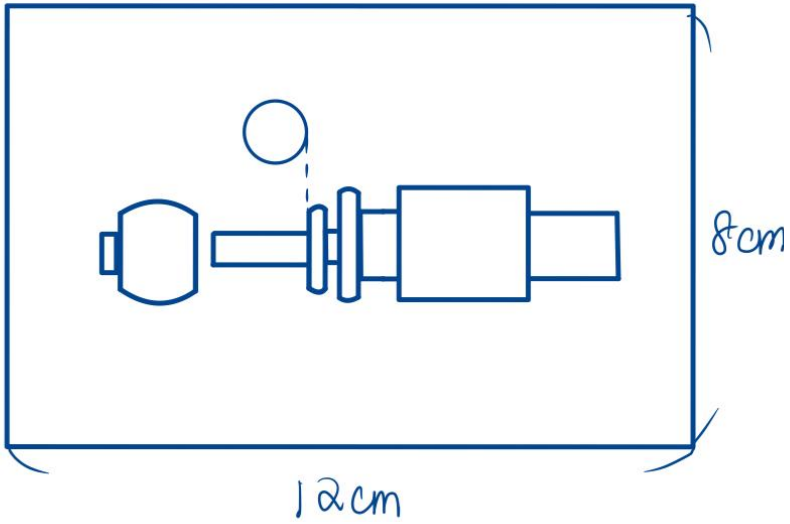
exciting coil

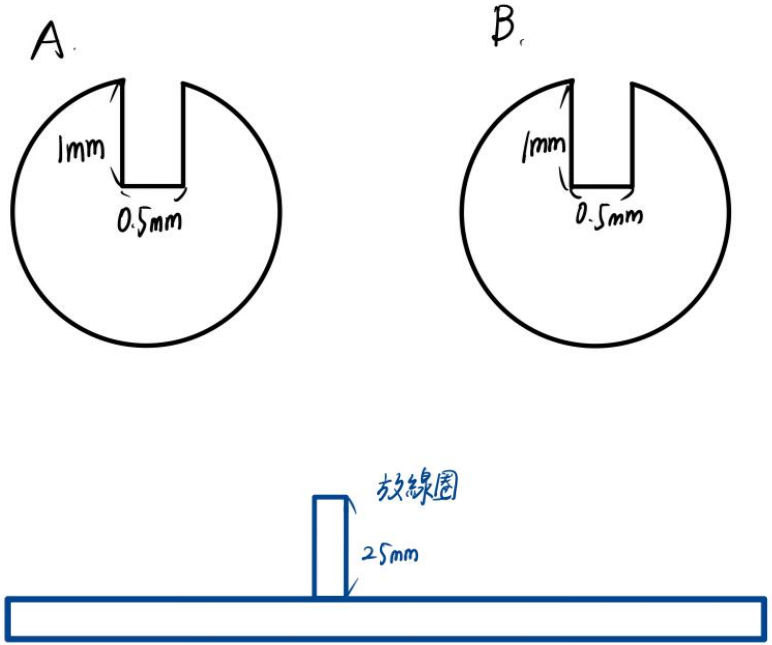






Final Design (Top View and Detail View)





Raw Materials

Diameter of the Insulated Copper Wire Used

15 μm

30 μm

40 μm

Bearing Specifications